1 The $G_{\alpha\rho}$-Gate: Empirical Falsifiability and β-Metric

1.1 Theoretical Formulation

We introduce the $G_{\alpha\rho}$-Gate as a formal mechanism to bridge the EABC-model predictions with observable astrophysical signatures. We assume a scalar-field-dependent coupling of the fine-structure constant α to the ambient vacuum density ρ. The effective coupling is defined by:

$$\alpha(\rho) = \alpha_0[1 + \beta \cdot f(\rho)] \tag{1}$$

where α_0 represents the laboratory-measured value, $\beta \simeq 7.9 \times 10^{-3}$ is the derived model constant, and $f(\rho)$ denotes the environmental screening function. We impose the following boundary conditions:

- **Local Screening:** $f(\rho_{\text{local}}) = 0$, ensuring consistency with terrestrial high-precision experiments and atomic clock constraints.
- **Asymptotic Limit:** $f(\rho \to 0) = 1$, defining the "Zero-Density-Limit" in the intergalactic medium (IGM).

1.2 Falsification Criteria ($A\rho$-Matrices)

The EABC-model predicts a relative shift in spectral lines, $\Delta\omega/\omega$, for metallic absorbers in low-density environments. Utilizing the Many-Multiplet (MM) method, the predicted sensitivity is governed by:

$$\frac{\Delta\omega}{\omega} \simeq K_i \cdot \beta \cdot f(\rho_{\text{void}}) \tag{2}$$

where K_i are the ionic sensitivity coefficients. Adopting $f(\rho_{\text{void}}) \approx 1$, the model yields the following predictions for selected ionic species:

Ion	K_i	$\Delta\omega/\omega$ **(Pred.)**
Mg II	0.05	$+3.95 \times 10^{-4}$
Fe II	0.09	$+7.11 \times 10^{-4}$
Cr II	0.15	$+1.19 \times 10^{-3}$
Zn II	0.22	$+1.74 \times 10^{-3}$

Table 1: Predicted spectral shifts for selected ions under the EABC-model assumption.

1.3 Discussion of the "Claim-Wall"

Current high-resolution spectroscopic observations (e.g., VLT/ESPRESSO) constrain relative variations of α to the order of $\sim 10^{-6}$. The EABC-model prediction of $\sim 10^{-3}$ significantly exceeds these observational limits.

We explicitly acknowledge this discrepancy not as a model failure, but as a robust Falsification-Gate. The EABC-model is thus uniquely defined by a "Claim-Wall": If subsequent deep-space spectroscopic analysis of low-density absorbers does not reveal a correlated shift in the specified range, the hypothesis of a universal α-coupling to the local vacuum density must be rejected. This operational approach shifts the discourse from vague phenomenological speculation to a hard, testable quantitative framework, inviting empirical verification or immediate refutation.